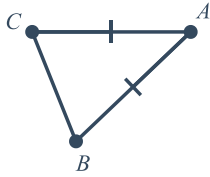


Line segments and distance

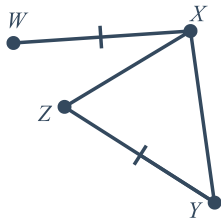


- 1 Draw the diagram that shows $\overline{EG}$ and a point F that is not on $\overline{EG}$.
- 2 The diagram shown has two congruent segments marked:



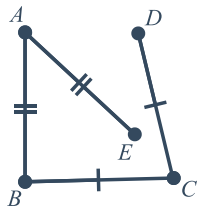
Identify which segment is equal in length to $\overline{AB}$.

- 3 The diagram shown has two congruent segments marked:



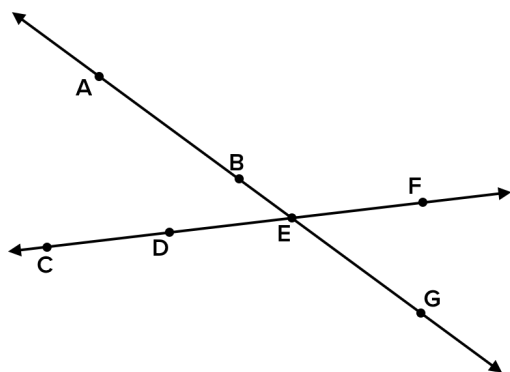
Identify which segment is equal in length to $\overline{ZY}$.

- 4 The diagram shown has two pairs of congruent segments marked:



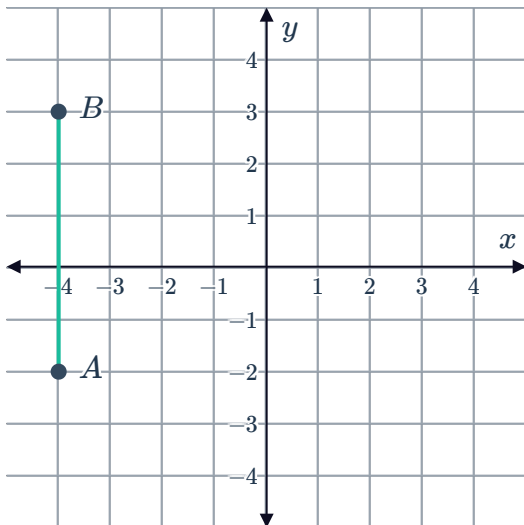
Identify which segment is equal in length to $\overline{AB}$.

5 What is another way to describe the exact set of points as $\overline{FD}$?

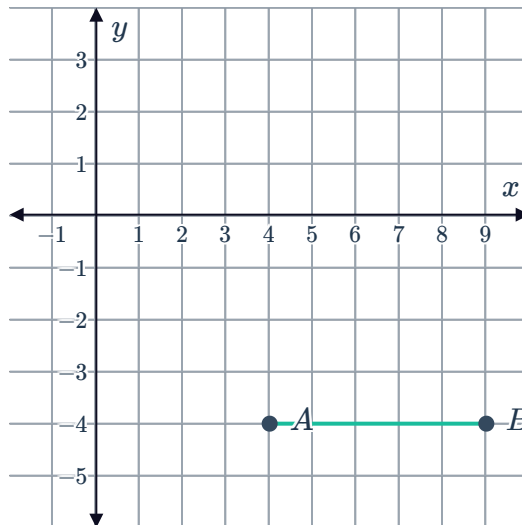


6 Find the length of the indicated segment.

a The vertical segment joining $A(-4, -2)$ and $B(-4, 3)$



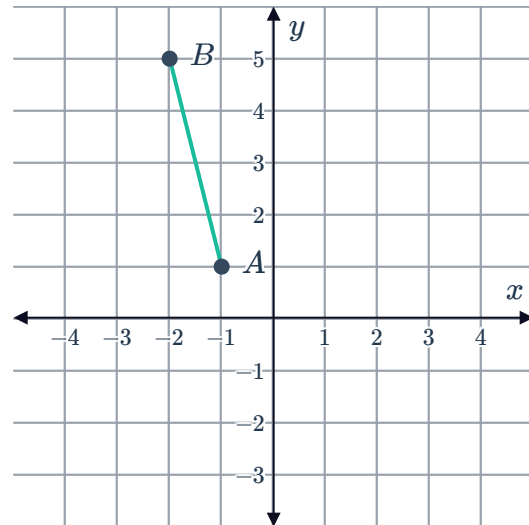
b The horizontal segment joining $A(4, -4)$ and $B(9, -4)$



7 Consider $\overline{AB}$ on the graph.

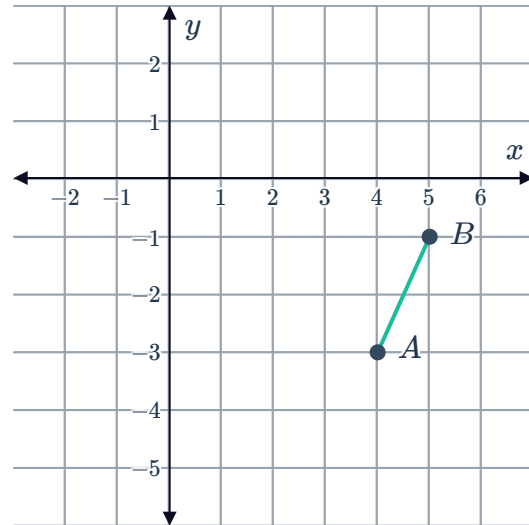


- a Identify the vertical component of the distance from A to B .
- b Identify the horizontal component of the distance from A to B .
- c If the length of $\overline{AB}$ is c , find the exact value of c .



8 Consider $\overline{AB}$ on the graph.

- a Identify the vertical component of the distance from A to B .
- b Identify the horizontal component of the distance from A to B .



9 State the formula for finding the distance d between two points (x_1, y_1) and (x_2, y_2) .

10 Find the missing values in the following working that is used to find the distance between points $(-5, 10)$ and $(4, -5)$.

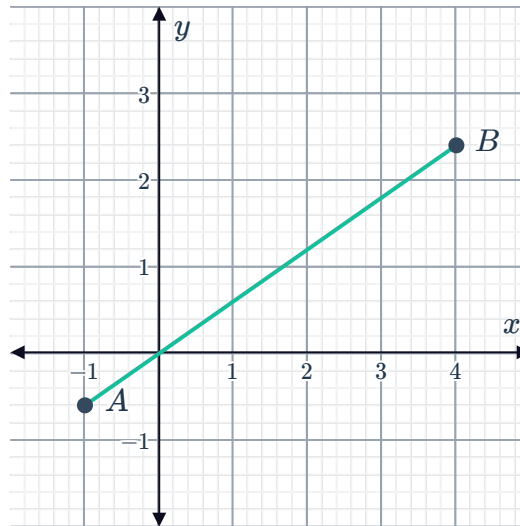
$$\text{Distance} = \sqrt{(-5 - \text{ })^2 + (10 - \text{ })^2}$$

11 Find the distance between $A(1, 4)$ and $B(7, 12)$, rounding your answer to two decimal places.

12 Find the exact distance between the following points.



- 13 Find the exact distance from $A\left(-1, -\frac{3}{5}\right)$ to $B\left(4, \frac{12}{5}\right)$.



- 14 Given that $M(-9, -1)$ and $N(1, 5)$.
- a Find the exact distance from M to the origin.
 - b Find the exact distance from N to the origin.
 - c Which point is closer to the origin?
- 15 Given that $P(5, -1)$, $M(10, -6)$, and $N(1, 2)$.
- a Find the exact distance from P to M .
 - b Find the distance from P to N .
 - c Which point is further from P ?
- 16 The points $M(0, -8)$ and $R(6, -2)$ are the end points of a segment. $N(5, -3)$ is a point on this segment.
- a Find the exact distance between M and N .
 - b Find the exact distance between N and R .
 - c In what ratio does N divide $\overline{MR}$?

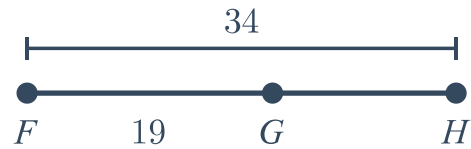
17 Find the length of the indicated segment.



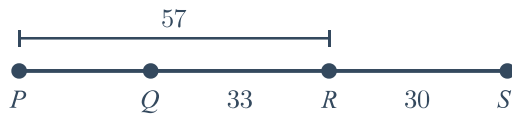
a $\overline{CE}$



b $\overline{GH}$



18 Consider the following figure.



a Find QS .

b Find PQ .

19 In the diagram, $JK = 5x + 2$, $KL = 7x + 4$, and $JL = 42$.



a Find x .

b Find KL .

20 In the diagram, $JT = 5x + 5$, $CT = 76$, and $CJ = 4x + 8$. Find JT .



21 For each of the following scenarios:

i Find x .

ii Find CB .

a Suppose the point B lies on $\overline{CD}$ between the points C and D , and $CB = 2x$, $BD = 4x$, and $BD = 12$.

b Suppose the point B lies on $\overline{CD}$ between the points C and D , and $CB = 3x - 6$, $BD = 4x + 3$, and $CD = 18$.

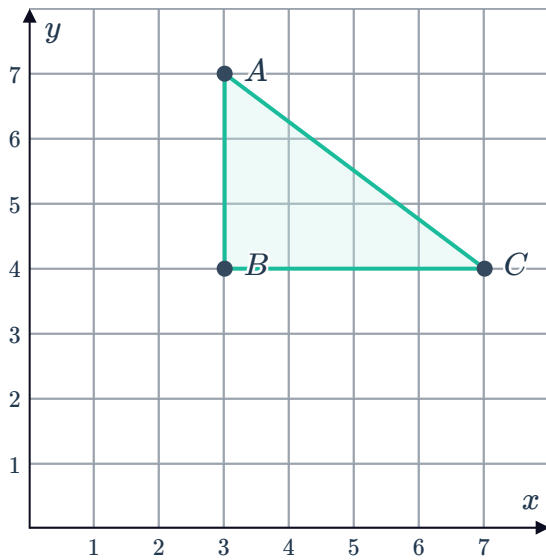


Additional questions

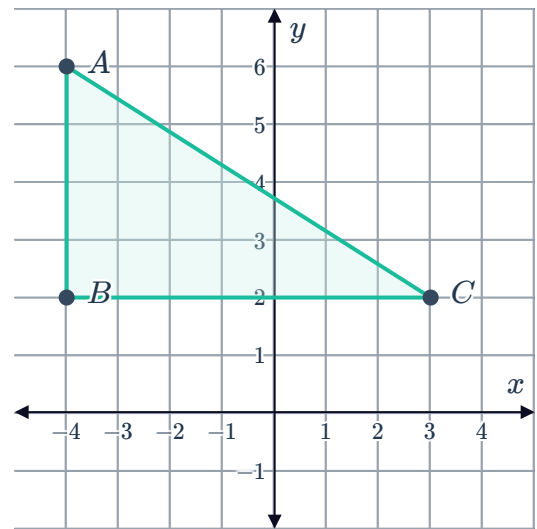
22 For each of the following right triangles:

- i Determine AB .
- ii Determine BC .
- iii Using the Pythagorean theorem, calculate AC , rounding your answer to two decimal places if necessary.

a



b

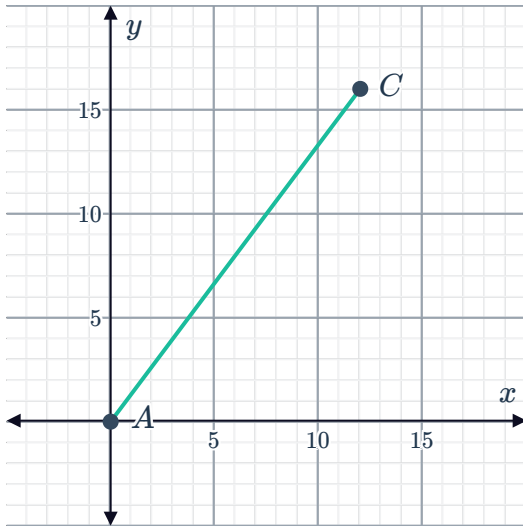


23 For each of the following coordinate planes

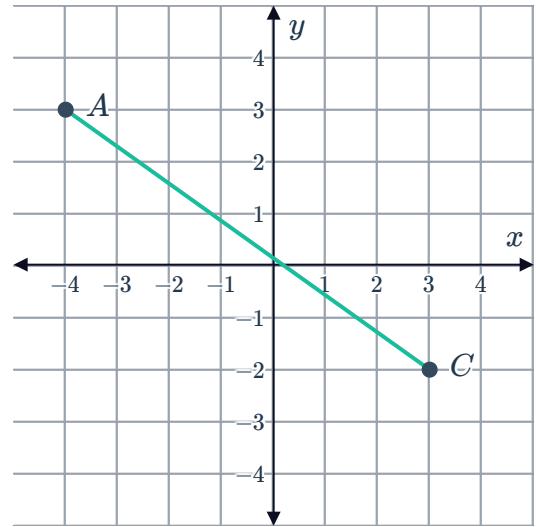


- i Copy the coordinate plane and draw a right triangle with $\overline{AC}$ as the hypotenuse.
- ii Label the shorter two sides with the measure of their lengths.
- iii Use the Pythagorean theorem to find the distance between A and C , round your answer to one decimal place if necessary.

a $A = (0, 0), C = (12, 16)$

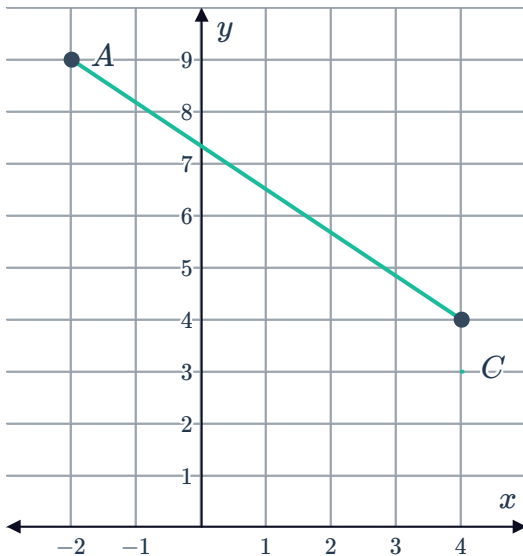


b $A = (-4, -3), C = (3, -2)$

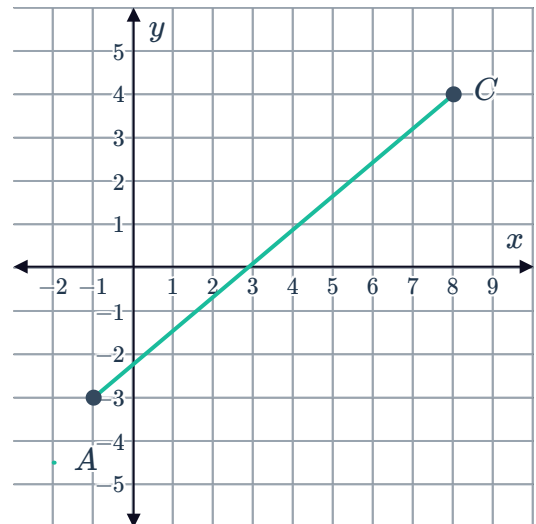


24 Consider $\overline{AC}$ that has been graphed on the coordinate plane.
Find AC , rounding your answer to two decimal places.

a



b



25 Find the distance between Point A and Point B , rounding your answer to two decimal places if necessary.



a $A(1, 4)$ and $B(7, 12)$

b $A(4, 2)$ and $B(-8, -7)$

c $A(-1, 9)$ and $B(-4, 1)$

d $A\left(-1, -\frac{3}{5}\right)$ and $B\left(4, \frac{12}{5}\right)$

26 Given $P(4, 3)$, $M(-3, -4)$ and $N(-7, 1)$:

a Find the distance from P to M , rounding your answer to two decimal places.

b Find the distance from P to N , rounding your answer to two decimal places.

c Determine the point that is further from P .

27 A circle with center at point $C(3, 4)$ has point $A(-12, 12)$ lying on its circumference. Find the radius of the circle.

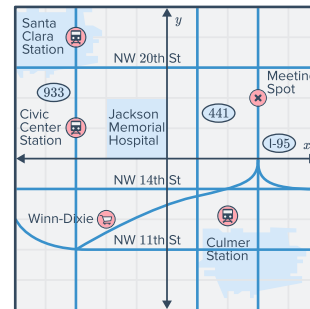
28 $ABCD$ is a rhombus whose vertices are $A(1, 2)$, $B(3, 10)$, $C(11, 12)$ and $D(9, 4)$. Find the exact length of the diagonals:

a $\overline{AC}$

b $\overline{BD}$

29 Mateo is meeting his mom at the Meeting Spot shown on the map below after a doctor's appointment. They need to catch the train home.

- a** Which of the three train stations is the closest to their meeting spot?
- b** What might not make the closest train station the best station to go to in reality?
- c** If they needed to stop at the Winn-Dixie before going home, which station would have the shortest trip distance?



30 $\overline{AM}$ is a vertical line segment which is 3 units long. If A is the point $(-2, 6)$:

- a** How many possible coordinates of M are there?
- b** What are the possible coordinates of M ?
- c** How would your answers to parts (a) and (b) change if we were not told the line segment was vertical?



- 32** Dayanna and David were asked for the formula for the distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ on the coordinate plane.

Dayanna wrote $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

David wrote $AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

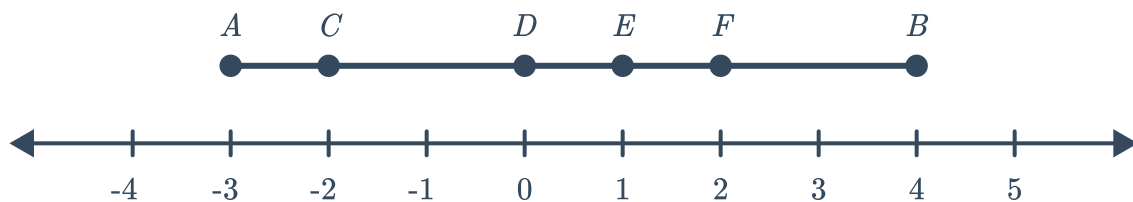
Who is correct? Explain your answer.

- 33** Fyfe is asked to find the length of the line segment $\overline{AB}$ with endpoints $A(-4, 7)$ and $B(3, -5)$. Below is their solution.

1	$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Distance formula
2	$= \sqrt{(3 - -5)^2 + (-4 - 7)^2}$	Substitution Property of Equality
3	$= \sqrt{(8)^2 + (-11)^2}$	Simplify
4	$= \sqrt{64 + 121}$	Simplify
5	$= \sqrt{185}$	Simplify

- a** Identify the step where Fyfe makes an error and explain their misunderstanding.
b Correct Fyfe's work.

- 34** Use the diagram to find each segment length:



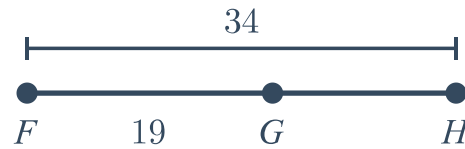
- | | |
|---------------|---------------|
| a AC | b AE |
| c EF | d FB |
| e DB | f AB |

35 Write an equivalence statement using the segment addition postulate, then find the length of indicated segment.

a CE

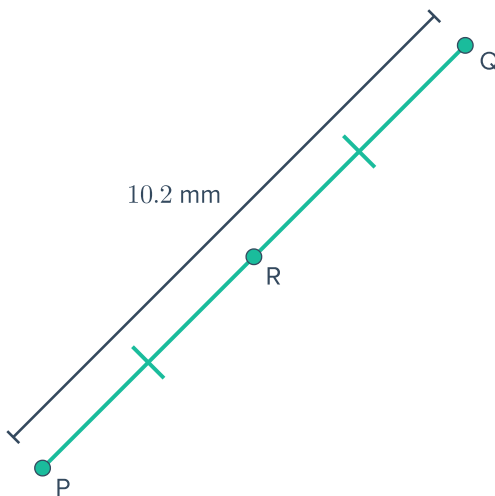


b GH

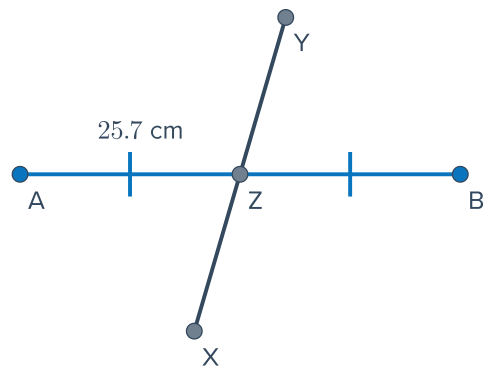


36 Use the diagrams provided to find the indicated length:

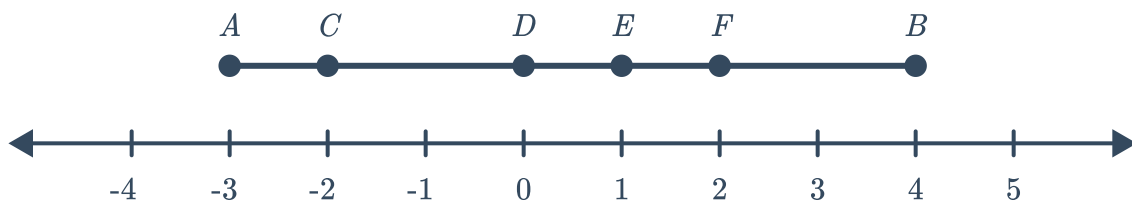
a PR



b AB



37 Consider the segment shown in the diagram:



- Which segments are congruent to $\overline{AC}$?
- Which segments are congruent to $\overline{AE}$?
- If a point G was added at -1 , how many additional congruent segments would you find in parts (a) and (b)?



38 Use the segment addition postulate to find the unknown length.

- a** Given: $JK = 5x + 2$, $KL = 7x + 4$ and $JL = 42$. Find KL .



- b** Given: $JT = 5x + 5$, $CT = 76$ and $CJ = 4x + 8$. Find JT .



39 Suppose the point B lies on the segment $\overline{CD}$ between the points C and D .

- a** If $CD = 4x$, $BD = 2x$ and $CB = 12$, what is the length of $\overline{BD}$? **b** If $CB = 3x - 6$, $BD = 4x + 3$ and $CD = 18$, what is the length of $\overline{CB}$?

40 In the diagram below, M is the midpoint of $\overline{KL}$. Find the length ML .



41 Given that points X , Y , and Z are collinear, and that $XY = 12$ and $XZ = 18$, what are the possible lengths for YZ ?

Draw diagrams that support your solution.

42 Given that $AB = 10$, $BC = 5$, and $AC = 12$ could points A , B , and C be collinear? Explain.